Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Whitmore, Mihriban Ph.D.		
Project Title:	Multipurpose Crew Restraints for Long Duration Space Flights		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-244-1004
Organization Name:	NASA - Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Holden, Kritina (Lockheed Martin)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	With permanent human presence onboard the International Space Station (ISS), a crew will be living and working in microgravity, interfacing with their physical environment. Without optimum restraints and mobility aids (R&MA's), the crewmembers may be handicapped for performing some of the on-orbit tasks. In addition to weightlessness, the confined nature of a spacecraft environment results in ergonomic challenges such as limited visibility and access to the activity area and may cause prolonged periods of unnatural postures. Thus, determining the right set of human factors requirements and providing an ergonomically designed environment are crucial to astronauts' well-being and productivity. The purpose of this project is to develop requirements and guidelines, and conceptual designs, for an ergonomically designed multi-purpose crew restraint. In order to achieve this goal, the project involves development of functional and human factors requirements, design concept prototype development, analytical and computer modeling evaluations of		

	concepts, two sets of microgravity evaluations and preparation of an implementation plan. It is anticipated that developing functional and design requirements for a multi-purpose restraint would facilitate development of optimum ergonomically designed restraints to accommodate the off-nominal but repetitive tasks, and minimize the performance degradation due to lack of optimum setup for onboard task performance. In addition, development of an ergonomically designed restraint concept prototype will allow verification and validation of the requirements defined.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The process to develop and refine Human Factors Engineering (HFE) requirements for crew restraints will be applicable to any HFE requirements generation effort within the hardware design and development cycle.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)